Han Yang

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Education

University of Massachusetts Amherst Sep. 2025 - Present
Ph.D. in Computer Science *Amherst, MA*

- Advisor: Prof. Chuang Gan
- Research Interest: Dexterous Manipulation, Robot Learning, Reinforcement Learning, Tactile Sensing

University of Pennsylvania Aug. 2023 - Dec. 2023
Exchange, School of Engineering and Applied Science *Philadelphia, PA*
GPA: **4.00**/4.00

The Chinese University of Hong Kong Sep. 2021 - May. 2025
B.Sc. in Computer Science, First-Class Honours *Hong Kong SAR, China*
GPA: **3.803**/4.000 (top 5%)

Publications & Manuscripts

(Equal contribution.)*

- **DexTaG: Tactile-as-Guidance in Reinforcement Learning for Dexterous Manipulation** **ICRA 2027 Submission**
  Han Yang, Yian Wang, Yunlong Song, Zhenjia Xu, and Chuang Gan.
  A tactile-guided RL method learns dexterous manipulation from human contact demonstrations and distills the resulting policy into a tactile-free controller.

- **SyncWorld: Visual Calibration Enables World Models as Zero-Shot Simulators** **NeurIPS 2026 Oral**
  Yuncong Yang*, Zhengtao Han*, Furkan Özyurt*, Zeyuan Yang, **Han Yang**, Junyi Cao, Haoyu Zhen, Yilun Du, and Chuang Gan.
  Introduced a visually calibrated world model that predicts the visual outcomes of numerical robot controls in unseen cameras, environments, and embodiments without downstream training.

- **PhyScensis: Physics-Augmented LLM Agents for Complex Physical Scene Generation** **ICLR 2026**
  Yian Wang*, **Han Yang***, Minghao Guo, Xiaowen Qiu, Tsun-Hsuan Wang, Wojciech Matusik, Joshua B. Tenenbaum, and Chuang Gan.
  Introduced an agentic framework that leverages procedural rules and a physics simulator to generate physically plausible and complex 3D scenes.

- **3D-Mem: 3D Scene Memory for Embodied Exploration and Reasoning** **CVPR 2025**
  Yuncong Yang*, **Han Yang***, Jiachen Zhou, Peihao Chen, Hongxin Zhang, Yilun Du, and Chuang Gan.
  Introduced a 3D scene memory framework that empowers VLM agents with lifelong memory, exploration, and reasoning abilities in 3D environments.

- **UniMuMo: Unified Text, Music, and Motion Generation** **AAAI 2025 Oral**
  Han Yang, Kun Su, Yutong Zhang, Jiaben Chen, Kaizhi Qian, Gaowen Liu, and Chuang Gan.
  Developed a unified generative model that seamlessly performs cross-modal generation between text, music, and human motion within a single, shared framework.

- **SILT: A Shadow-aware Iterative Label Tuning Approach for Learning to Detect Shadows from Noisy Labels** **ICCV 2023**
  Han Yang*, Tianyu Wang*, Xiaowei Hu, and Chi-Wing Fu.
  Proposed a novel iterative label tuning approach that leverages shadow-aware physical properties to progressively refine noisy annotations and train robust shadow detection models.

Experience

- **Analog Devices Inc. — Robot Learning Intern**
  May 2026 – Sep. 2026
  - Developed a **shared-autonomy framework for dexterous-hand teleoperation**, inferring object-level manipulation intent from human hand motion and scene context.
  - Trained **reinforcement-learning policies** to assist challenging dexterous behaviors such as in-hand translation and reorientation while preserving the teleoperator's intended motion.
  - Investigated how tactile feedback can improve dexterous manipulation and shared-autonomy control.
- **Genesis AI — Robot Learning Engineer Intern**
  Nov. 2025 – Jan. 2026
  - Built and integrated **dexterous-hand teleoperation infrastructure** for robot learning and manipulation data collection.
  - Developed an egocentric glove-based manipulation data-collection pipeline with tactile sensing for VLA training.
  - Collaborated on **DexTaG**, a tactile-guided reinforcement learning framework for learning dexterous tool-use behaviors from human contact demonstrations.
- **MIT-IBM Watson AI Lab — Research Assistant Intern**
  May 2023 – Feb. 2025
  - Conducted research spanning **multimodal learning, generative modeling, and embodied AI**, working closely with Prof. Yilun Du and Prof. Chuang Gan.
  - Developed learning-based methods for multimodal generation and embodied reasoning, contributing to projects including **UniMuMo** and **3D-Mem**.

Awards & Honors

- Yasumoto International Exchange Scholarship (2023, top 1 in ongoing exchange students)
- Arthur and Louise May Memorial Scholarship (2023, top 1 in CS major);
- Professor Omar Wing Memorial Scholarship (2022, top 1 in CS major)
- Award for Outstanding Academic Performance (2023-2025, top 5% in CS major)
- Dean's List (2022-2025, top 10% in the department)

Skills

Programming	Python, C/C++
ML	PyTorch, Reinforcement Learning, Imitation Learning, VLA/VLM
Robotics/Simulation	Isaac Lab, Isaac Sim, MuJoCo, Genesis, ROS/ROS2
Hardware	WUJI, Tesollo DG-5F, xArm, Franka, Manus
Other	SolidWorks, Blender